

IC 113: Complex and Vector Calculus

Course Code	IC 113
Course Name	Complex and Vector Calculus
L-T-P-C	(1.5-0.5-0-2)
Prerequisites	Math 1
Intended for	B. Tech. 1st Year
Mutual Exclusion	(None)
Approval	50th BoA (Jul 2023); 44th Senate (Sep 2024), in the B.Tech (all branches) - Institute Core curriculum

Course Contents

- **Complex variable:** Limit, continuity, differentiability and analyticity of functions, Cauchy-Riemann equations, line integrals in complex plane, Cauchy integral theorem, independence of path, existence of indefinite integral, Cauchy's integral formula, derivatives of analytic functions, Taylor's series, Laurent's series, Zeros and singularities, Residue theorem, evaluation of real integral [10 Lectures]
- **Integration:** Riemann integral, Double integral and its applications, Fubini's theorem, Volumes and Areas, Change of variable in double integral. Special cases: Polar coordinates, Triple integral, Applications, Change of variable in triple integral. Special cases: Cylindrical and Spherical coordinates, Surface area, Surface integral, Line integrals, Green's theorem, Vector fields Divergence and Curl of a vector field, Stoke's theorem, The divergence theorem. [11 Lectures]

Text Books

1. Thomas and Finney, **Calculus and Analytical Geometry**, 9th Edition, Addison and Wesley Publishing Company, 1996.
2. R. V. Churchill and J. W. Brown, **Complex Variables and Applications**, 9th Editions, 2021.

References

1. E. Kreyszig, **Advanced Engineering Mathematics**, 10th Edition.
2. S. Ponnusamy, **Foundations of Complex Analysis**, 2nd Edition, Narosa, 1995.
3. J. E. Marsden, A. J. Tromba and A. Weinstein, **Basic Multivariable Calculus**, Springer, 1993.